

Unit 2.5 Limit at infinity §

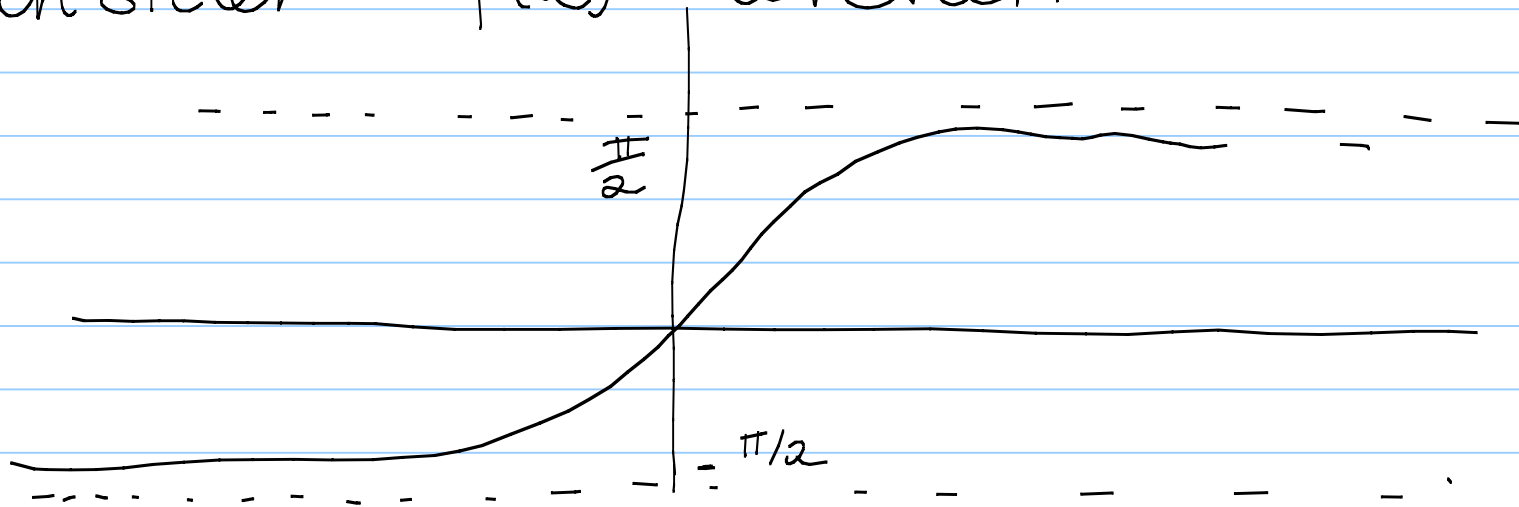
Note Title

2012/03/18

horizontal asymptotes.

Section 2.6 in Stewart p130-143.

Example
Consider $f(x) = \arctan x$



What is $\lim_{x \rightarrow \infty} \arctan x = \pi/2$

$\lim_{x \rightarrow -\infty} \arctan x = -\pi/2$

Def Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means The values of $f(x)$ can be made arbitrary close to L by taking x sufficiently large.

Def Let f be defined on $(-\infty, a)$

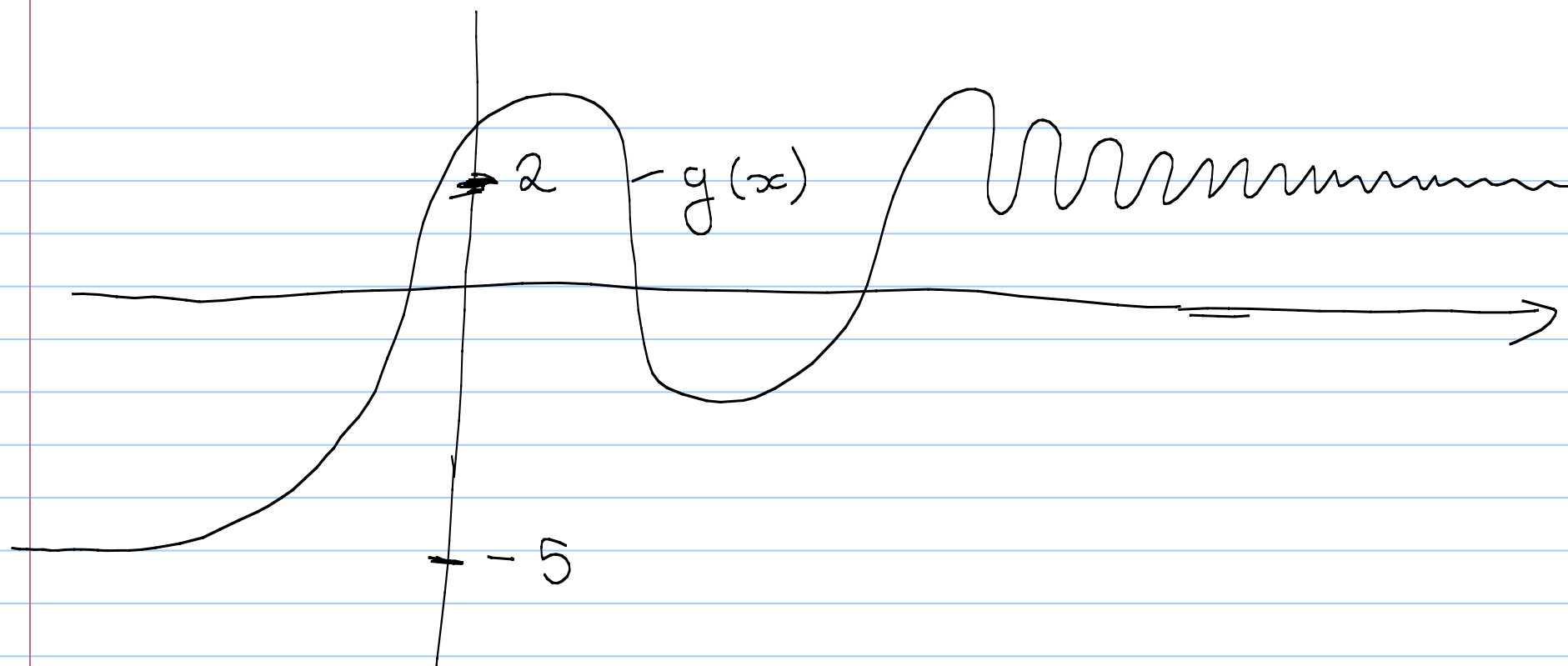
$$\lim_{x \rightarrow -\infty} f(x) = L$$

means The values of $f(x)$ can be made arbitrary close to L by taking x sufficiently large negative

Example



$$\lim_{x \rightarrow \infty} f(x) = 2$$



$$\lim_{x \rightarrow \infty} g(x) = 2$$

$$\lim_{x \rightarrow -\infty} g(x) = -5$$

Def The line $y = L$ is called
a horizontal asymptote of the
curve $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Example What are the horizontal asymptotes of $f(x) = \arctan x$?

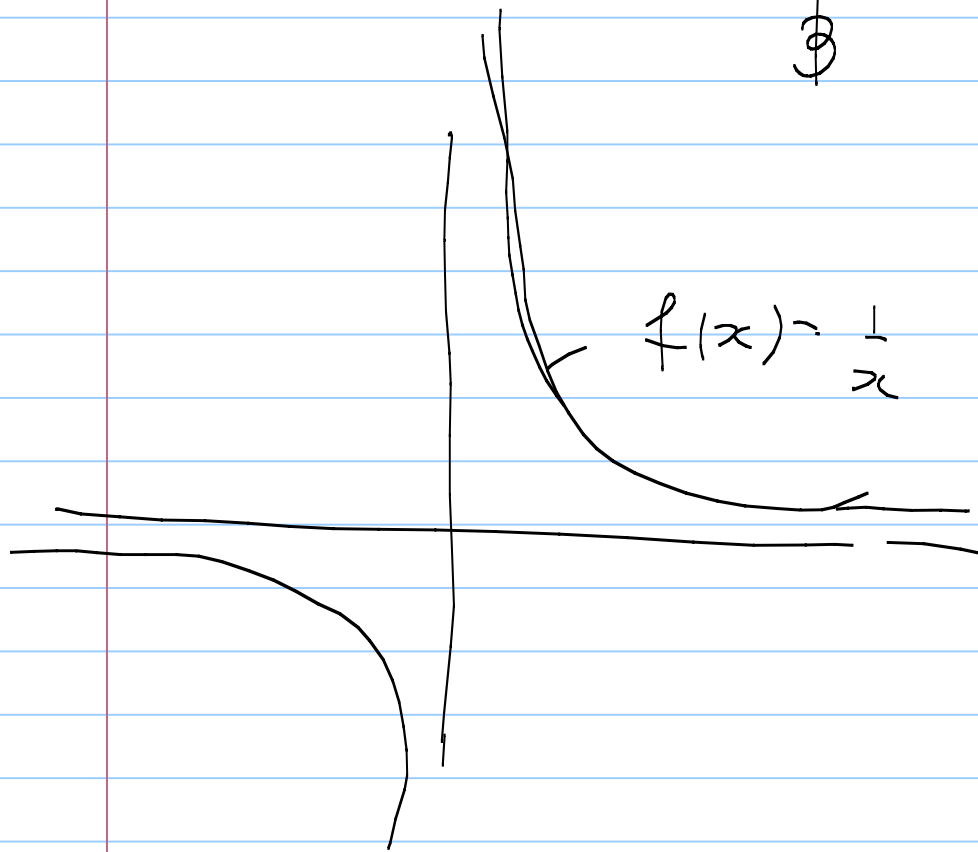
$$y = \pi/2$$

and $y = -\pi/2$

Example Find $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

§

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$



$$\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{1}{x^4} = 0$$

Theorem If $r > 0$ is a rational number, then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

• If $r > 0$ is a rational number such that x^r is defined for all x then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

Example

$$1] \lim_{x \rightarrow \infty} \frac{1}{x^{100}} = 0$$

$$2] \lim_{x \rightarrow -\infty} \frac{1}{x^9} = 0$$

When finding infinite limits of rational functions of the form

$$f(x) = \frac{Q(x)}{P(x)}$$

we know what $Q(x)$ & $P(x)$ are polynomials

The answer will be, BUT

You have to show the

Calculations!

If $f(x) = \frac{Q(x)}{P(x)}$

and

$Q(x)$ & $P(x)$ are polynomials

- degree of $Q(x) = \text{degree of } P(x)$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{Q(x)}{P(x)} = \text{real number}$$

- degree of $Q(x) < \text{degree of } P(x)$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{Q(x)}{P(x)} = 0$$

• degree of $Q(x) >$ degree of $P(x)$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{Q(x)}{P(x)} = \infty \text{ or } -\infty$$

Similar if $x \rightarrow -\infty$

Example Find horizontal asymptotes,
if any of

$$1) f(x) = \frac{2x^4 - x^3}{4x^4 + 2}$$

FORM
 $\left[\frac{\infty}{\infty} \right]$

$$\lim_{x \rightarrow \infty} \frac{2x^4 - x^3}{4x^4 + 2} = \frac{2 \lim_{x \rightarrow \infty} x^4 - \lim_{x \rightarrow \infty} x^3}{4 \lim_{x \rightarrow \infty} x^4 + 2}$$

$$= \lim_{x \rightarrow \infty} \frac{x^4 (2 - \frac{1}{x})}{x^4 (4 + \frac{2}{x^4})}$$

$$= \lim_{x \rightarrow \infty} \frac{2 - \frac{1}{x}}{4 + \frac{2}{x^4}}$$

$$= \frac{2 - \lim_{x \rightarrow \infty} \frac{1}{x}}{4 + \lim_{x \rightarrow \infty} \frac{2}{x^4}} = \frac{2}{4} \quad \left| \begin{array}{l} \text{line} \\ y = \frac{1}{2} \end{array} \right. \rightarrow$$

$$2] \quad f(x) = \frac{4x^3 - 3x^2 + 2x}{8x^4 + 2x^2 - 1}$$

$$\lim_{x \rightarrow \infty}$$

$$\frac{4x^3 - 3x^2 + 2x}{8x^4 + 2x^2 - 1}$$

FORM
 $\left[\frac{\infty}{\infty} \right]$

$$= \lim_{x \rightarrow \infty}$$

$$\frac{x^4 \left(\frac{4}{x} - \frac{3}{x^2} + \frac{2}{x^3} \right)}{x^4 \left(8 + \frac{2}{x^2} - \frac{1}{x^4} \right)}$$

$$= \frac{\lim_{x \rightarrow \infty} \frac{4}{x} - \lim_{x \rightarrow \infty} \frac{3}{x^2} + \lim_{x \rightarrow \infty} \frac{2}{x^3}}{8 + \lim_{x \rightarrow \infty} \frac{2}{x^2} - \lim_{x \rightarrow \infty} \frac{1}{x^4}}$$

$$= \frac{0 - 0 + 0}{8 + 0 - 0} = 0$$

$y = 0$ is The Horizontal asymptote

$$3) \quad f(x) = \frac{x^3 - x^2 + 7}{8x^2 + 7x}$$

$$\lim_{x \rightarrow \infty} \frac{x^3 - x^2 + 7}{8x^2 + 7x}$$

FARIM

$\left[\frac{\infty}{\infty} \right]$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x}^2 (x - 1 + \frac{7}{x^2})}{\cancel{x}^2 (8 + 7/x)} = \infty$$

↓

Recall.

PA7

$$\sqrt{x^2} = |x|$$

This means

$$\sqrt{x^2} = x \quad \text{if } x \geq 0$$

$$\sqrt{x^2} = -x \quad \text{if } x < 0$$

Example Find horizontal asymptotes of

$$f(x) = \frac{\sqrt{9x^2 - 4}}{4 - 8x}$$

$$= \frac{\sqrt{x^2 (9 - 4/x^2)}}{x (\frac{4}{x} - 8)}$$

$$= \frac{\sqrt{x^2} \sqrt{9 - 4/x^2}}{x(4/x - 8)}$$

$$= \frac{|x| \sqrt{9 - 4/x^2}}{x(4/x - 8)}$$

$$\lim_{x \rightarrow \infty} \frac{x \sqrt{9 - 4/x^2}}{x(4/x - 8)} = \frac{\sqrt{9 - \lim_{x \rightarrow \infty} \frac{4}{x^2}}}{\lim_{x \rightarrow \infty} \frac{4}{x} - 8}$$

$$= \frac{3}{-8}$$

$$\lim_{x \rightarrow -\infty} \frac{-x \sqrt{9 - 4/x^2}}{x(4/x - 8)} = \frac{-3}{-8} = \frac{3}{8}$$

There are TWO horizontal asym

$$y = -\frac{3}{8} \quad \text{and} \quad y = \frac{3}{8}$$

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Example. Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$

$$f(x) = \frac{2 - 4x^3 + 16x^4}{2x^4 + 3x}$$

$$\lim_{x \rightarrow \infty} \frac{2 - 4x^3 + 16x^4}{2x^4 + 3x}$$

$$= \lim_{x \rightarrow \infty} \frac{x^4 \left(\frac{2}{x^4} - \frac{4}{x} + 16 \right)}{x^4 \left(2 + 3/x^3 \right)} = \frac{16}{2} = 8$$

line $y = 8$ is a horizontal asymptote

Example, Find horizontal asym.

$$\square \quad f(x) = \frac{x}{\sqrt{1+x^2}}$$

$$= \frac{x}{\sqrt{x^2 \left(\frac{1}{x^2} + 1 \right)}}$$

$$= \frac{x}{\sqrt{x^2} \sqrt{\frac{1}{x^2} + 1}}$$

$$= \frac{x}{|x| \sqrt{\frac{1}{x^2} + 1}}$$

$$\sqrt{x^2} = |x|$$

$$\sqrt{x^2} = x \text{ if } x > 0$$

$$\sqrt{x^2} = -x \text{ if } x < 0$$

$$\lim_{x \rightarrow \infty} \frac{x}{x \sqrt{\frac{1}{x^2} + 1}}$$

$$= \frac{1}{\sqrt{\lim_{x \rightarrow \infty} \frac{1}{x^2} + 1}} = \underline{1}$$

$$\lim_{x \rightarrow -\infty} \frac{x}{-x \sqrt{\frac{1}{x^2} + 1}} = -1$$

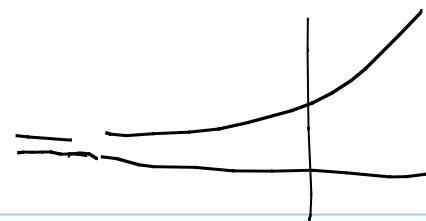
$y = 1$ and $y = -1$ are horizontal asymptotes.

$$2] \quad h(x) = \frac{\sqrt{4x^6 + 9x^2}}{3x^3 + 2x^2 - 4x}$$

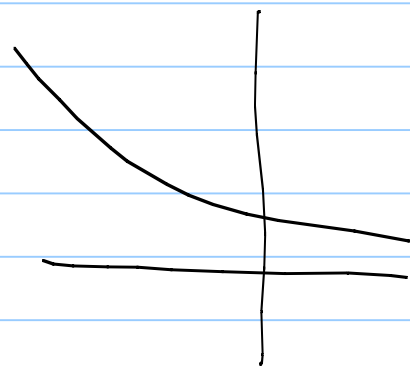
The line $y = \frac{2}{3}$ & $y = -\frac{2}{3}$
horizontal asym

$$\lim_{x \rightarrow \infty} \frac{x^3 \sqrt{4 + 9/x^4}}{x^3 (3 + 2/x - 4/x^2)} = \frac{\sqrt{4}}{3} = \frac{2}{3}$$

Example Find



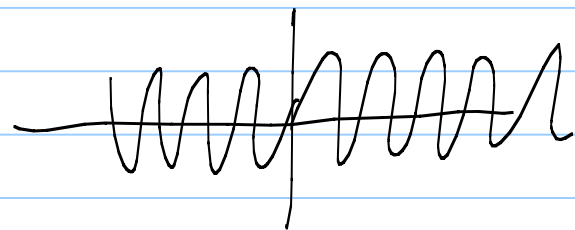
$$\lim_{x \rightarrow -\infty} e^x = 0$$



$$\lim_{x \rightarrow \infty} e^{-x} = 0$$



$$\lim_{x \rightarrow \infty} \sin x = \text{Does not exist.}$$



Example

Find horizontal asymptotes, if any of

$$1] f(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$\lim_{x \rightarrow -\infty} \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$= \frac{\lim_{x \rightarrow -\infty} e^{2x} - \lim_{x \rightarrow -\infty} 1}{\lim_{x \rightarrow -\infty} e^{2x} + \lim_{x \rightarrow -\infty} 1} = \frac{0 - 1}{0 + 1} = -1$$

$$\lim_{x \rightarrow \infty} \frac{e^{2x} - 1}{e^{2x} + 1}$$

NO
 $\left(\frac{\infty}{\infty} \right) = 1$

$$= \lim_{x \rightarrow \infty} \frac{e^{2x} - 1}{e^{2x} + 1} \left(\frac{e^{-2x}}{e^{-2x}} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{1 - e^{-2x}}{1 + e^{-2x}}$$

$$= 1.$$

$$2] \quad f(x) = \frac{2 + \cos x}{x^2}$$

$$-1 \leq \cos x \leq 1$$

$$1 \leq \cos x + 2 \leq 3$$

$$\lim_{x \rightarrow \infty} x^2 = \infty$$

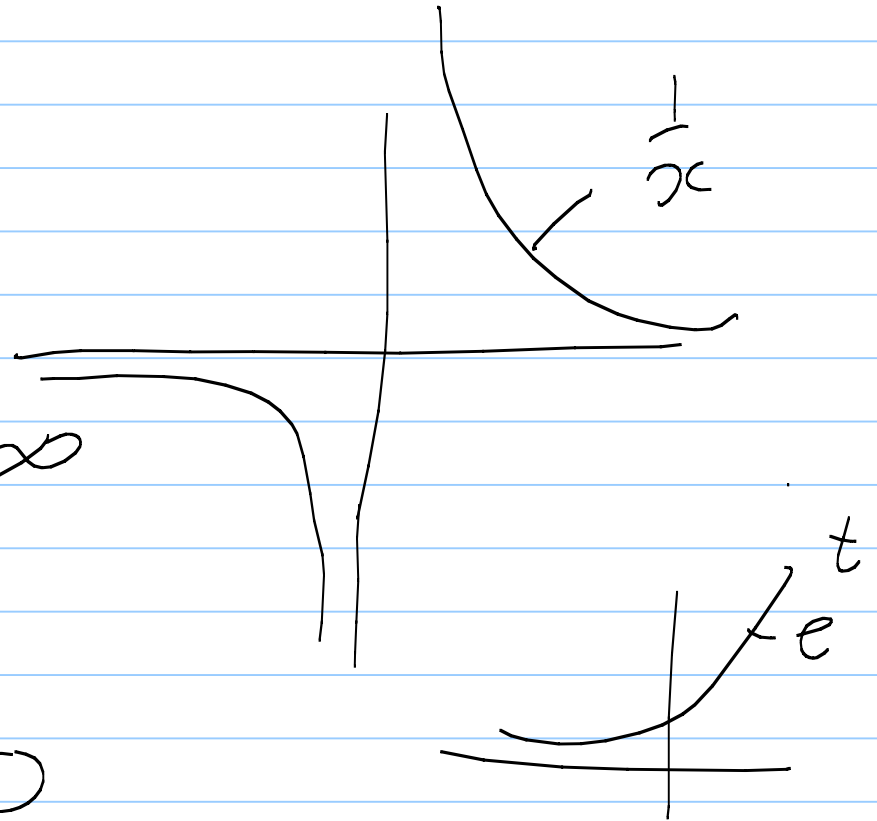
$$\therefore \lim_{x \rightarrow \infty} \frac{2 + \cos x}{x^2} = 0$$

Evaluate $\lim_{x \rightarrow 0^-} e^{\frac{1}{x}}$ NB

let $t = \frac{1}{x}$

$x \rightarrow 0^-$, $t \rightarrow -\infty$

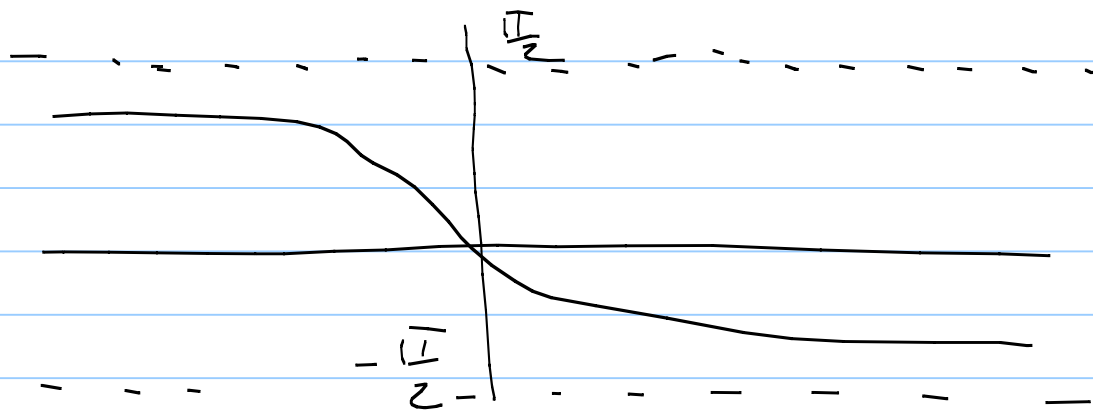
$\lim_{t \rightarrow -\infty} e^t = 0$



You need to be able to use graphs to find horizontal asymptotes

Example Find horizontal asym.

1] $f(x) = \arctan(-2x^3)$

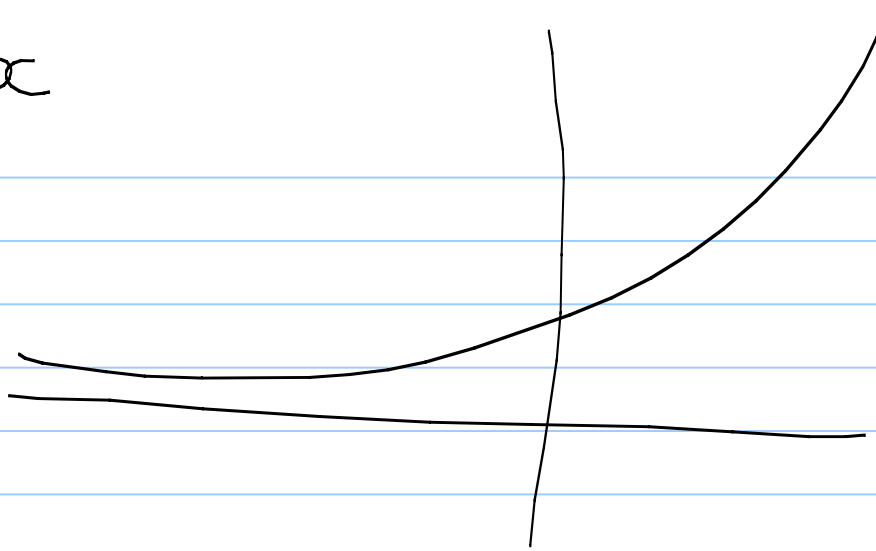


$$y = \frac{\pi}{2} \text{ and}$$

$$y = -\frac{\pi}{2}$$

horizontal asymp.

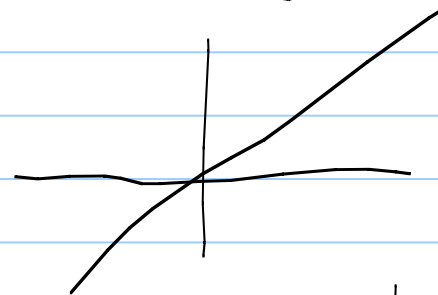
$$f(x) = e^{3x}$$



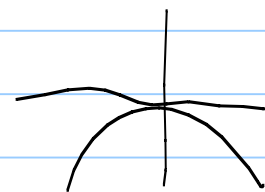
$$y = 0$$

Infinite Limits at infinity

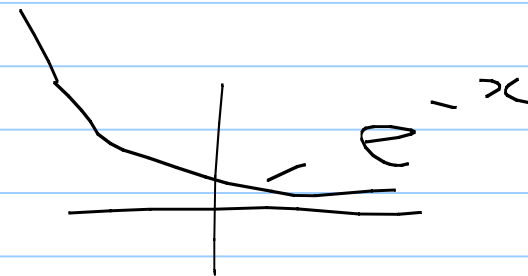
$$\bullet \lim_{x \rightarrow \infty} f(x) = \infty$$



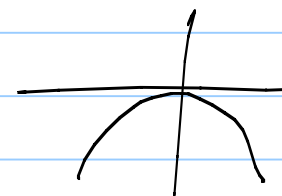
$$\bullet \lim_{x \rightarrow \infty} f(x) = -\infty$$



$$\bullet \lim_{x \rightarrow -\infty} f(x) = \infty$$

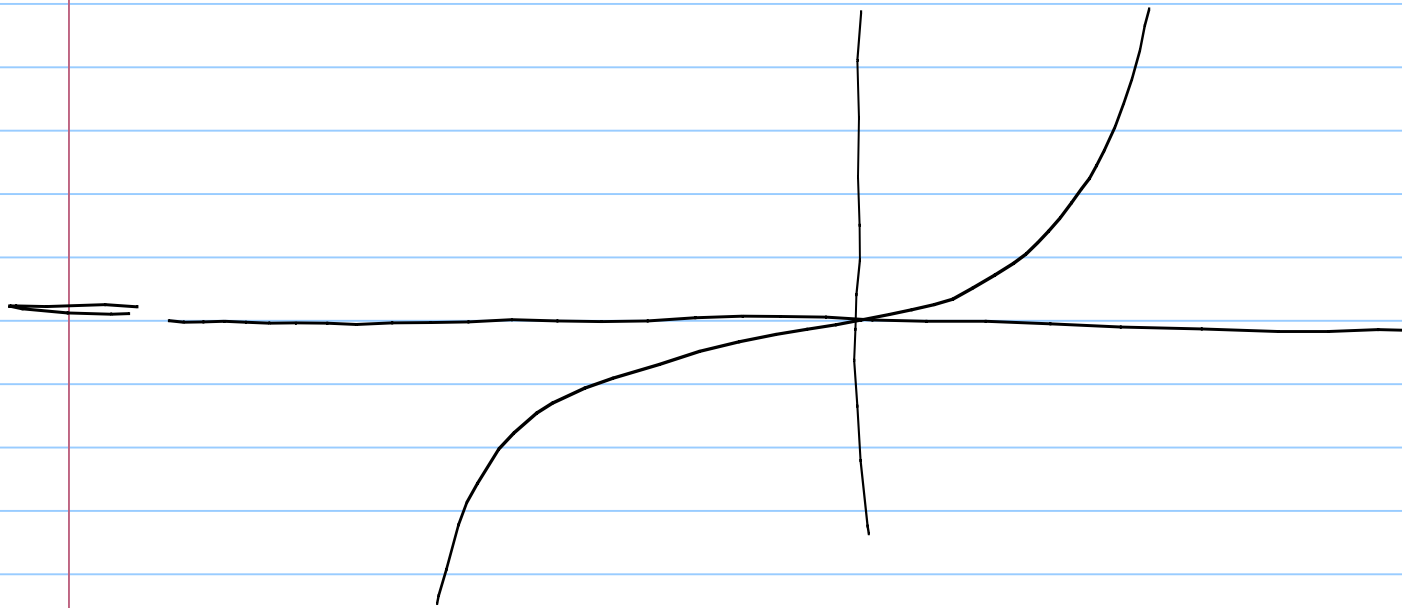


$$\bullet \lim_{x \rightarrow -\infty} f(x) = -\infty$$



Examples

$$f(x) = x^3$$



$$\lim_{x \rightarrow \infty} f(x) = \infty$$

$$\lim_{x \rightarrow -\infty} f(x) = -\infty$$

Example Find

$$\lim_{x \rightarrow \infty} (x^2 - x)$$

$$= \lim_{x \rightarrow \infty} x(x - 1)$$

$$= \lim_{x \rightarrow \infty} x \lim_{x \rightarrow \infty} (x - 1)$$

$$= \infty$$

